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Los Alamos, New Mexico

June 2, 2026

Voyager Technologies
Advanced Technologies Group

Dear Hiring Manager,

With a Ph.D. in Computational Mechanics and extensive experience developing machine learning surrogate models for nuclear reactor materials, I am excited to apply for the Material Scientist position at Voyager Technologies. My expertise in data-driven materials modeling, combined with hands-on experimental characterization capabilities, aligns directly with your requirements for advancing materials research through computational and ML-based approaches.

At Los Alamos National Laboratory, I developed the "LAROMance" surrogate model for polycrystalline nuclear materials by curating datasets from thousands of low-length-scale simulations and applying machine learning methods to bridge microstructure to engineering-scale predictions. This work on metal alloys (HT-9, Grade 91 steel) under extreme conditions reached R&D 100 Award Finals twice. My Ph.D. research established an integrated computational-experimental framework combining finite element modeling, custom piezoelectronic instrumentation with feedback control, and advanced microscopy (SEM, AFM, profilometry) with sample preparation (sputter coating) to characterize adhesion in flexible microelectronic systems—resulting in five peer-reviewed publications.

I am particularly drawn to Voyager's mission to advance space and defense technologies. Translating my nuclear materials modeling expertise to space applications, while leveraging my experience integrating LLM-based tooling into engineering workflows, represents an exciting opportunity to contribute to mission-critical materials challenges.

I would welcome the opportunity to discuss how my computational materials science background and experimental characterization skills can support Voyager's materials research initiatives.

Sincerely,

Andre Ruybalid